

UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI



MACHINE LEARNING AND DATA MINING II
LABWORK 1: PCA

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I. Study the dataset

1.1. Choose 2 datasets from Kaggle or any source in Internet with more than 6 dimensions.

For this exercise, we choose:

Dataset 1: Breast Cancer Wisconsin (Diagnostic) Dataset:

The dataset consists of 570 rows with 569 samples. While the original data contains 32 dimensions and 30 features, this labwork focuses on the first 10 dimensions including 8 selected features (from diagnosis to concave points_mean).

Dataset 2 Wine Quality Dataset:

The dataset contains 1144 rows with 1143 samples. It spans 13 dimensions with 12 features.

1.2. For each dataset, determine which feature is discrete or continuous? quantitative or qualitative? numerical or categorical? Explain.

For Breast Cancer dataset:

All 8 features of first 10 dimensions are Continuous, Quantitative and Numerical (ratio) bc they are float represent the results of measurements and calculations of the physical characteristics of the cell (radius measurement, area calculation, roughness measurement...). Because they are actual measurement results, they are real numbers (with decimal parts) and can take any value within a continuous range (radius could be 14.1, 14.12, or 14.125...), therefore they are continuous, quantitative and numerical data.

*(For diagnosis: This is the **Label (Target)**, not a feature. It is **Categorical, Qualitative, and Discrete (Nominal)**)*

For wine quality dataset:

All features are continuous and quantitative because they are float represent the results of physicochemical measurements of the wine's properties. Because they are actual measurement results, they are real numbers (with decimal parts) and can take any value within a continuous range (fixed acidity could be 7.4, 7.42, or 7.425...), therefore they are continuous, quantitative and numerical data.”

*(For quality: This is the **Label (Target)**, not a feature. It is **Numerical, Quantitative, and Discrete (Ordinal)**)*

1.3. What are the issues lying within data? for example: missing data, incorrect data? Do you need to prepare data?

For Breast Cancer dataset

a) Issues lying within data:

- Missing Data: There are no missing values in these first 10 dimensions.
- Incorrect Data / Formats (Dữ liệu chưa phù hợp)
 - `Diagnosis` column: It contains string values ('M', 'B'), being used as labels. Machine learning models generally require numerical inputs, so this format is invalid for direct model feeding then must be converted to **0** and **1** for **binary classification**
 - `id` column: Using it as a feature would be incorrect because the model might learn spurious patterns from the **ID** numbers.
- Duplicates: There are no duplicate rows within this subset (first 10 dimensions) of data.

b) Do you need to prepare data?

The preparation is needed:

- **Remove Irrelevant Features:** Drop the `id` column. It is just an identifier and provides no predictive power for the diagnosis.
- **Encoding:** Convert the `diagnosis` column from text ('M'/'B') to numbers (1,0) so it can be used as the target variable for training.
- **Scaling:** The numerical features have different scales (`area\_mean` is in the hundreds/thousands, while `smoothness\_mean` is < 0.2).

Scaling (**Standardization** or **Normalization**) is recommended to ensure fair weight distribution in many algorithms.

For Wine Quality dataset

a) Issues lying within data:

There are no missing and incorrect values

b) Do you need to prepare data?

- **Feature Scaling:** The numerical features exhibit significantly different scales. For instance, total sulfur dioxide reaches a maximum of 289.0, while chlorides and density remain below 1.0. Therefore, **Standardization (StandardScaler)** is mandatory to normalize the data to a mean of 0 and a standard deviation of 1, especially before applying PCA.
- **Remove Irrelevant Features:** Drop the 'id' column as it is a metadata identifier and provides no predictive value.
- **Data Cleaning:** No further action is required for null values or duplicates.
- **Encoding:** Not required, as all remaining input features are already in numeric format.

1.4. Is there any label in the datasets? Explain

For Breast Cancer dataset: The dataset contains a label.

In the context of Supervised Learning, the "label" is the specific outcome the model aims to predict. For this dataset, the objective is to determine whether a breast mass is cancerous based on its physical characteristics.

The diagnosis column provides the ground truth for each sample:

- **M (Malignant):** Cancerous/Harmful.
- **B (Benign):** Non-cancerous/Harmless.

While all other columns (such as radius, texture, ...) serve as features (inputs), the diagnosis serves as the target (output) that the model learns to identify.

For Wine quality dataset

In the context of **Supervised Learning**, the quality column serves as the label, where the objective is to predict a wine's quality score based on its chemical properties. Since the label consists of discrete ratings ranging from 3 to 8, this represents a **multi-class classification** task rather than a simple binary one.

*(Note: **Data Imbalance**: A key observation is that the classes are imbalanced. Most samples fall into the mid-range scores (5 and 6), while extreme scores like 3 or 8 are much rarer. This imbalance should be carefully considered during model training to ensure the model does not become biased toward the majority classes.)*

1.5 Statistical Analysis Results

Mathematical formulas:

Mean:
$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Sample Variance:
$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Sample Covariance:
$$cov_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Pearson Correlation:
$$r = \frac{cov_{xy}}{s_x s_y}$$

Calculation method:

Since the datasets have several samples, the statistical measures were calculated using the **Pandas** library in **Python**. Specifically, the `.mean()`, `.var()`, `.cov()`, and `.corr()` functions were applied to the numerical features to ensure accuracy and efficiency.

For Breast Cancer dataset

	radius	texture	perimeter	area	smoothness	compactness	concavity	concave point
Mean	14.1273	19,2896	91.969	654.889	0.0964	0.1043	0.0888	0.0489
Variance	12.4189	18.4989	590.441	123844	0.0002	0.0028	0.0064	0.0015

Table 1: Mean & Variance of Breast Cancer Dataset

	radius	texture	perimeter	area	smoothness	compactness	concavity	concave point
radius	12.4189	4.9076	85.4471	1224.48	0.0085	0.0942	0.1901	0.1125
texture	4.9076	18.4989	34.4389	485.994	-0.0014	0.0538	0.1037	0.049
perimeter	85.4471	34.4398	590.441	8435.77	0.0708	0.7147	1.3872	0.8024
area	1224.48	485.994	8435.77	123844	0.8762	9.2649	19.2449	11.242
smoothness	0.0085	-0.0014	0.0708	0.8762	0.0002	0.0005	0.0006	0.0003
compactness	0.0942	0.0538	0.7147	9.2649	0.0005	0.0028	0.0037	0.0017
concavity	0.1901	0.1037	1.3872	19.2449	0.0006	0.0037	0.0064	0.0029
concave point	0.1125	0.049	0.8024	11.242	0.0003	0.0017	0.0029	0.0015

Table 2: Covariance Matrix of Breast Cancer Dataset

	radius	texture	perimeter	area	smoothness	compactness	concavity	concave point
radius	1	0.3228	0.9879	0.9874	0.1706	0.5061	0.6768	0.8225
texture	0.3228	1	0.3295	0.3211	-0.0234	0.2367	0.3024	0.2935
perimeter	0.9979	0.3295	1	0.9865	0.2073	0.5569	0.7161	0.851
area	0.9874	0.3211	0.9865	1	0.177	0.4985	0.686	0.8233
smoothness	0.1706	-0.0234	0.2073	0.177	1	0.6591	0.522	0.5537
compactness	0.5061	0.2367	0.5569	0.4985	0.6591	1	0.8831	0.8311
concavity	0.6768	0.3024	0.7161	0.686	0.522	0.8831	1	0.9214
concave point	0.8225	0.2935	0.851	0.8233	0.5537	0.8311	0.9214	1

Table 3: Correlation Matrix of Breast Cancer Dataset

For wine dataset:

	fixed	volatile	citric	residual	chlorides	free sulfur	total sulfur	density
Mean	8.31111	0.531339	0.268364	2.53215	0.0869326	15.6155	45.9147	0.99673
Variance	3.05409	0.0322681	0.0386853	1.83851	0.0022342	105.072	1074.67	0.00000370588

Table 4: Mean and Variance of Wine Dataset

	fixed	volatile	citric	residual	chlorides	free sulfur	total sulfur	density
fixed acidity	3.05409	-0.07871	0.231382	0.40716	0.0089120	-2.95273	-6.33789	0.00229
volatile acidity	-0.07871	0.03226	-0.01922	-0.0014	0.0004783	-0.00361357	0.457838	0,00000571
citric acid	0.23138	-0.01922	0.038685	0.04688	0.0022806	-0.116107	0.237737	0.00014
residual sugar	0.40716	-0.00140	0.046888	1.83851	0.0045416	2.29801	8.48059	0.00099
chlorides	0.00891	0.00047	0.002280	0.00454	0.0022342	0.00740358	0.07463	0,000019
free sulfur dioxide	-2.95273	-0.00361	-0.11610	2.29801	0.0074035	105.072	222.149	-0.00106
total sulfur dioxide	-6.33789	0.45783	0.237737	8.48059	0.07463	222.149	1074.67	0.00316
density	0.00229	0,00000571	0.000142	0.00099	0,000019	-0.00106854	0.00316642	0,00000371

Table 5: Covariance Matrix of Wine Dataset

	fixed	volatile	citric	residual	chlorides	free sulfur	total sulfur	density
fixed acidity	1	-0.256123	0.673157	0.171831	0.107889	-0.154031	-0.110603	0.681501
volatile acidity	-0.256123	1	-0.544187	-0.0057511	0.0563363	-0.00196248	0.0777477	0.0165115
citric acid	0.673157	-0.544187	1	0.175815	0.245312	-0.0575891	0.0368711	0.375243
residual sugar	0.171831	-0.0057511	0.175815	1	0.0708631	0.165339	0.190379	0.380147
chlorides	0.107889	0.0563363	0.245312	0.0708631	1	0.0152935	0.0481633	0.208901
free sulfur dioxide	-0.154031	-0.00196248	-0.0575891	0.165339	0.0152935	1	0.66106	-0.05415
total sulfur dioxide	-0.110603	0.0777477	0.0368711	0.190379	0.0481633	0.66106	1	0.0501748
density	0.681501	0.0165115	0.375243	0.380147	0.208901	-0.05415	0.0501748	1

Table 6: Correlation Matrix of Wine Dataset

1.6. Most Correlated Features Analysis

For Breast Cancer dataset

In the Breast Cancer dataset, the most significant correlation occurs between `radius_mean` and `perimeter_mean`, reaching an exceptionally high coefficient of 0.9979. This near-perfect positive relationship is scientifically expected due to the geometric nature of the cell nuclei. Since these cells are roughly circular, their perimeter is mathematically proportional to their radius according to the formula.

From a machine learning perspective, this result indicates a high level of **Multicollinearity**. Because both features effectively convey the same information regarding the "size" of the tumor, they are considered redundant. Therefore, to simplify the model and improve training efficiency, we can safely drop one of these features—such as `perimeter_mean` without losing any critical predictive power.

For Wine Quality dataset

For the Wine Quality dataset, the strongest relationship is found between fixed acidity and pH, showing a negative correlation of -0.6852. This result is highly consistent with chemical principles, as the pH scale measures the concentration of hydrogen ions; as the concentration of acid (fixed acidity) increases, the pH level naturally decreases.

It is important to note that while this is the strongest pair in the Wine dataset, its coefficient is significantly lower than the geometric correlation found in the Breast Cancer data. This suggests that the chemical properties of wine possess more independent variation and less redundancy. However, other notable correlations exist, such as the positive relationship between citric acid and fixed acidity (0.67), confirming that specific acids are major contributors to the overall acidity profile of the wine.

1.7 Is there any missing data? How do you handle this issue

No, there were no missing data on both datasets

II. PCA

2.1. Principal Component Analysis (PCA) Selection

For Breast Cancer dataset

1. How do you select the principal components?

For Breast Cancer Dataset

PCA was performed on the 8 numeric features to reduce dimensionality while preserving data integrity. Using a 95% **Cumulative Explained Variance** threshold, we determined that 4 components are optimal, effectively halving the dimensions from 8 $\rightarrow$ 4. Just 3 components capture approximately 95% of the total variance, while 4 components retain 98.2%, demonstrating significant data redundancy in the original features.

To validate this selection, we also applied the **Elbow Method** and **Kaiser's Rule**. The **Elbow Method** identifies the elbow point on a scree plot where adding more components yields diminishing returns in variance. Simultaneously, **Kaiser's Rule** suggests retaining only components with an eigenvalue greater than 1.0, ensuring each selected component explains more variance than a single original feature.

Based on these criteria, 4 components were selected: this choice satisfies the 95% variance threshold (achieving 98.2%), aligns with the elbow point visible after PC3-PC4 on the scree plot, and includes components with the highest explanatory power. These criteria together ensure a balanced and efficient dimensionality reduction.

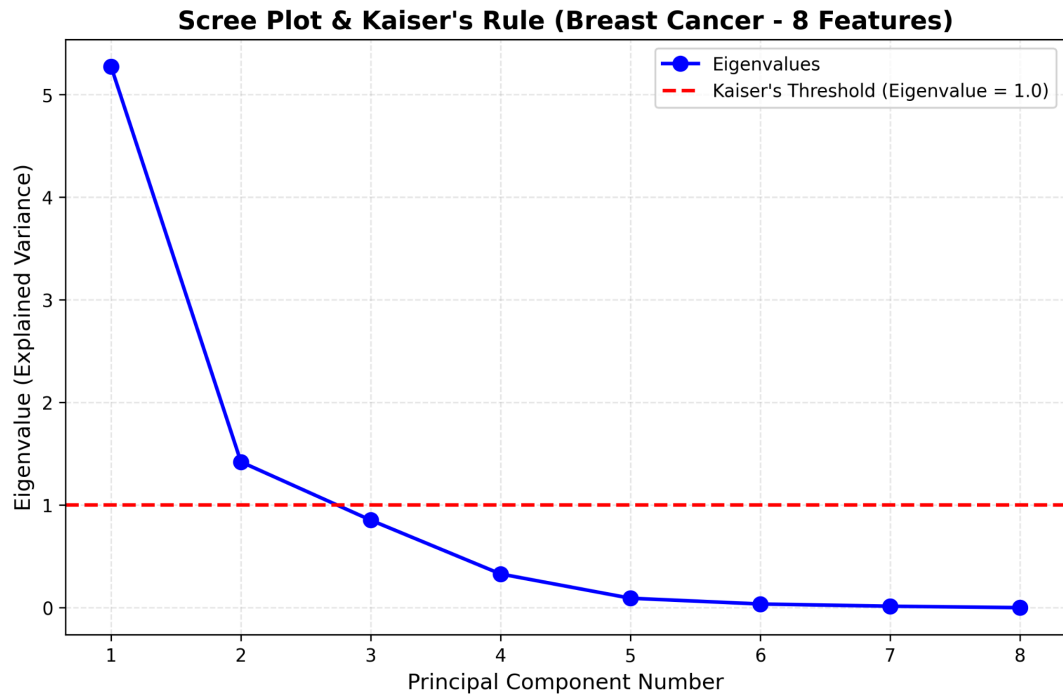


Figure 1: Scree Plot & Kaiser's Rule (Breast Cancer - 8 Features)

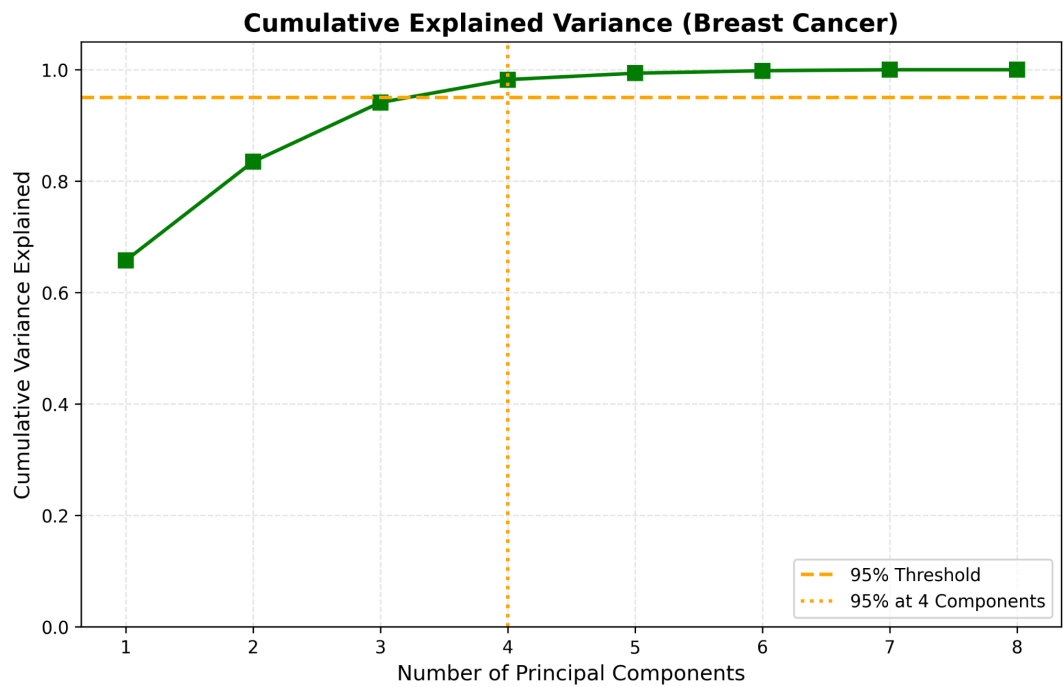


Figure 2: Cumulative Explained Variance (Breast Cancer-8 Features)

For Wine Quality dataset

PCA was performed on the 11 numeric features to reduce dimensionality while preserving data integrity. Using a 95% Cumulative Explained Variance threshold, we determined that 9 components are required, reducing dimensions from $11 \rightarrow 9$. Just 8 components capture approximately 94.7% of the total variance, while 9 components retain 97.9%, showing moderate data redundancy compared to other datasets.

To validate this selection, we also applied the “Elbow Method” and “Kaiser's Rule”. The Elbow Method identifies the "elbow" point on a scree plot where adding more components yields diminishing returns in variance. Simultaneously, Kaiser's Rule suggests retaining only components with an eigenvalue greater than 1.0, ensuring each selected component explains more variance than a single original feature.

Based on these criteria, 9 components were selected: this choice satisfies the 95% variance threshold (achieving 97.9%), aligns with the gradual decline visible on the scree plot, and retains components with meaningful explanatory power. These criteria together ensure a balanced and efficient dimensionality reduction.

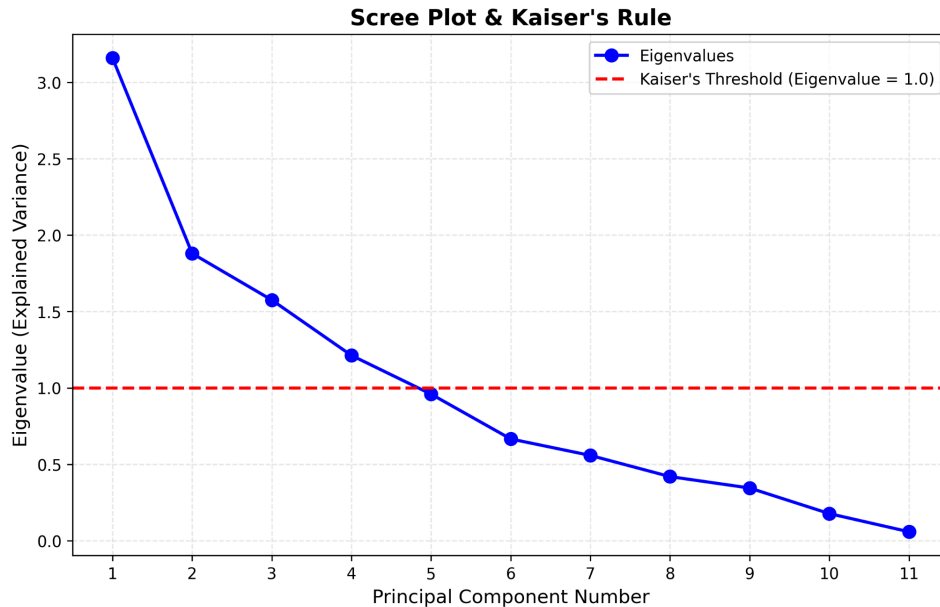


Figure 3: Scree Plot & Kaiser's Rule (Wine Quality Dataset)

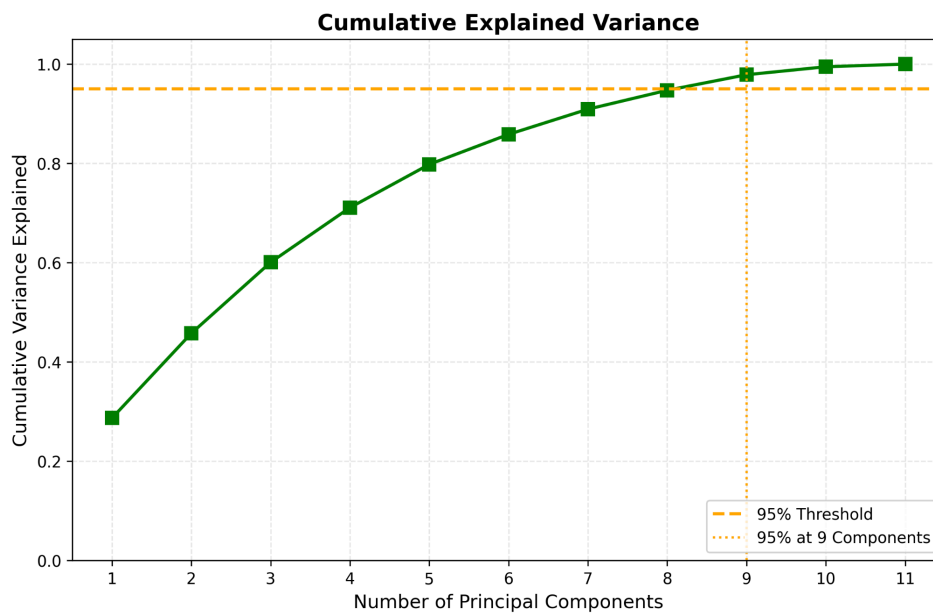


Figure 4: Cumulative Explained Variance (Wine Quality Dataset)

2.2. Is there any difference while using principal components with highest and lowest values?

For Breast Cancer Dataset

YES, there is a fundamental difference.

- **High Signal Content:** Even with just these 8 features, our **PCA** shows that they contain significant, structured variability (Outcome: 83% variance in just 2D). This implies strong patterns exist.
- **Core Features:** These 8 features represent the Mean values (Radius, Texture, Area, etc.). In medical imaging, the "Mean" (average size/texture) is the most predictive signal.

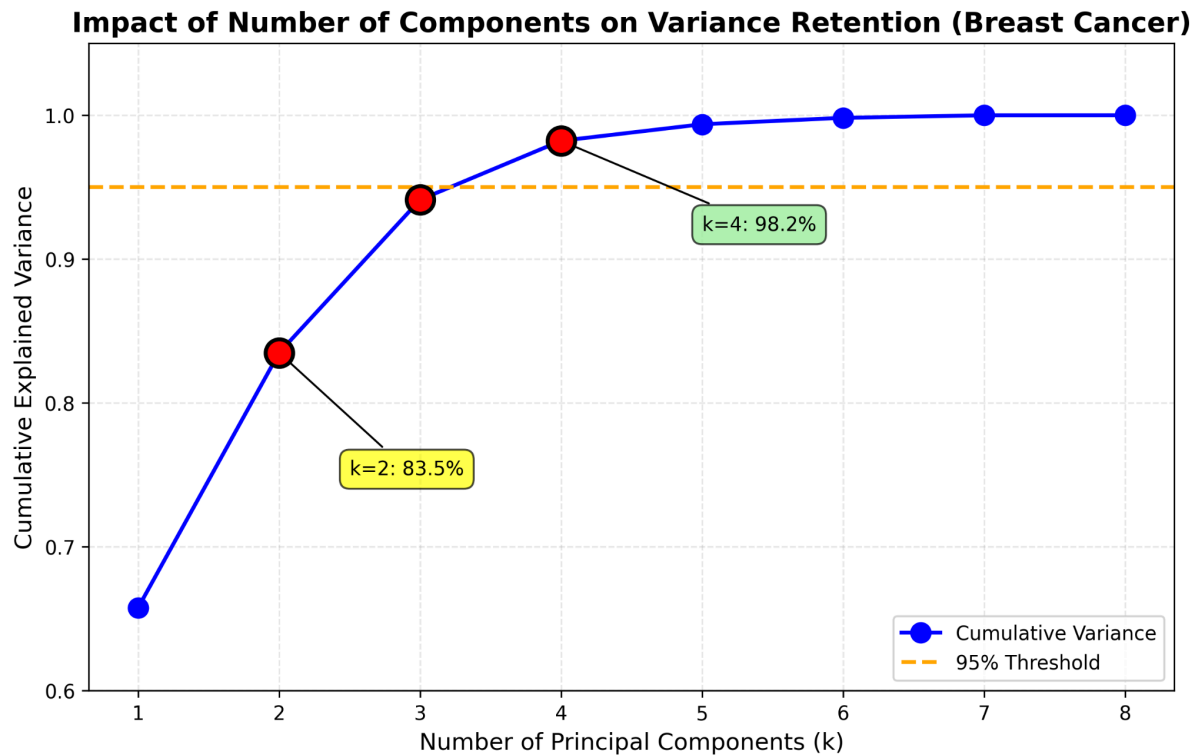


Figure 5: Number of Components on Variance Retention(Breast Cancer Dataset)

For Wine Quality Dataset

Yes, but with important caveates

- **Moderate Signal Content:** Our PCA analysis reveals that Wine features are less redundant than Breast Cancer. Even with 2 components, we only capture 45.8% variance (vs. 83% for Breast Cancer). This suggests the features are more independent and each carries unique information.
- **Complex Feature Space:** These 11 features represent distinct physicochemical properties (acidity, sugar, alcohol, sulfur dioxide, etc.). Unlike medical imaging where "mean" values dominate, wine quality depends on a delicate balance across multiple dimensions. No single feature or small subset is overwhelmingly predictive.

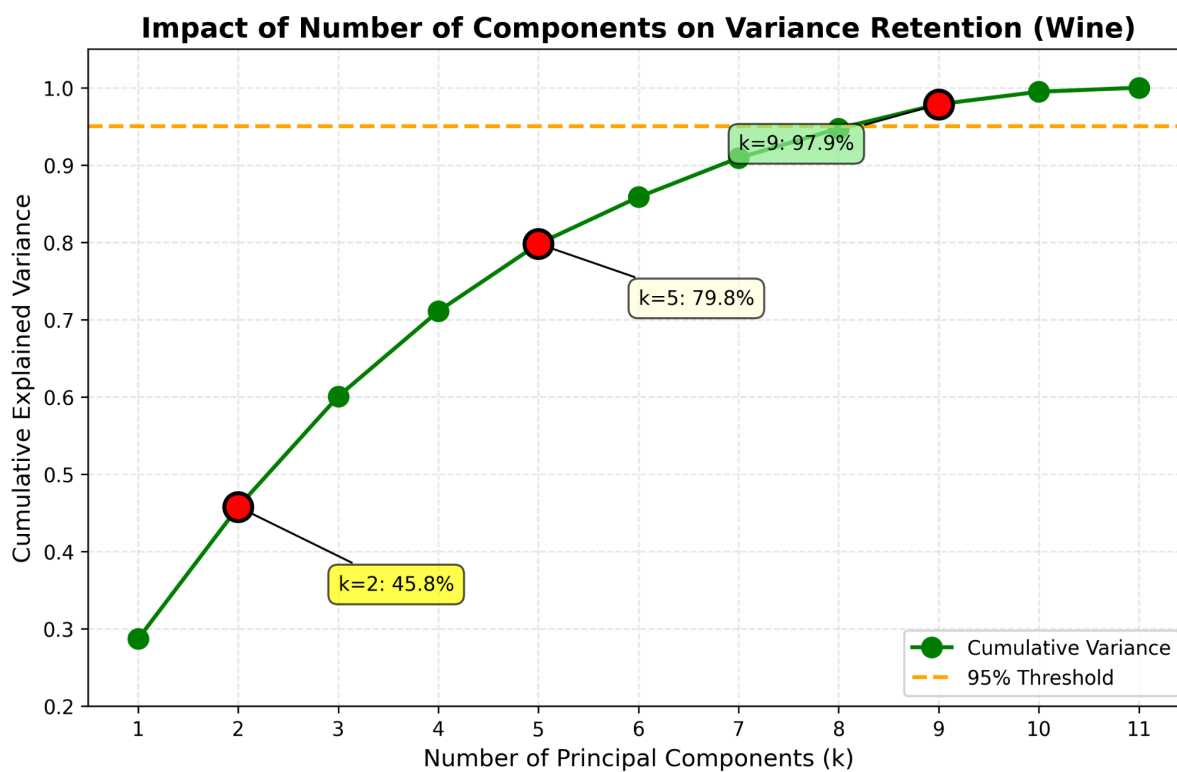


Figure 6: Number of Components on Variance Retention (Wine Quality Dataset)

2.3. Visualize data distribution in 2D. For Wine Quality Dataset:

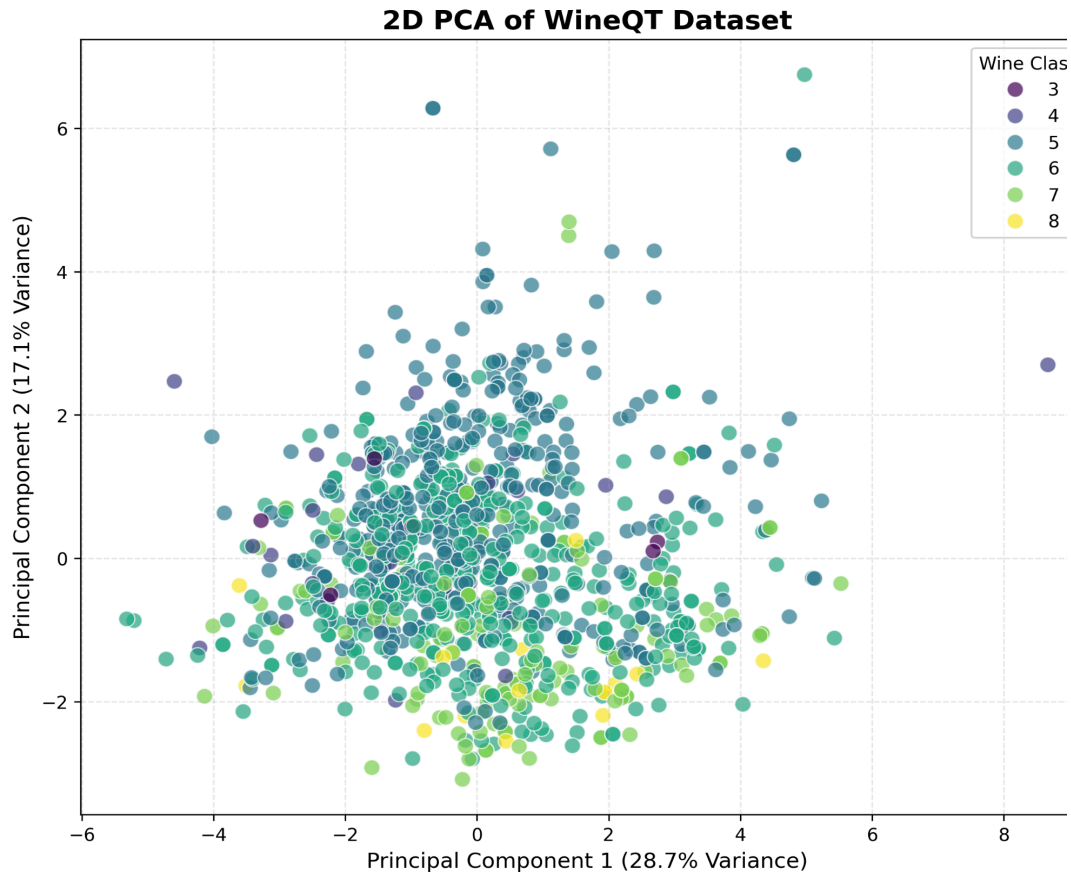


Figure 7: PCA 2D of Wine Quality Dataset

Analysis of Results

Severe Class Overlap

Unlike the **Breast Cancer** dataset (which showed clear separation), the Wine quality classes heavily overlap in 2D space. Points from different quality scores (3-8) are intermixed, making visual distinction nearly impossible.

Information Loss

This severe overlap directly reflects the low variance retention (45.8%) discussed in Section 2.2. Compressing 11 complex dimensions into 2D discards over 54% of the information, destroying much of the class structure.

Problem Complexity

Multi-class challenge: Predicting wine quality (6 ordinal classes: 3-8) is inherently harder than binary classification (Malignant/Benign).

Subtle boundaries: The chemical difference between a quality-5 wine and a quality-6 wine is very subtle and cannot be captured by just 2 principal components.

Non-linear relationships: Wine quality may depend on complex, non-linear interactions between features (e.g., the balance between acidity and sweetness), which PCA's linear projection cannot fully represent.

For Breast Cancer Dataset

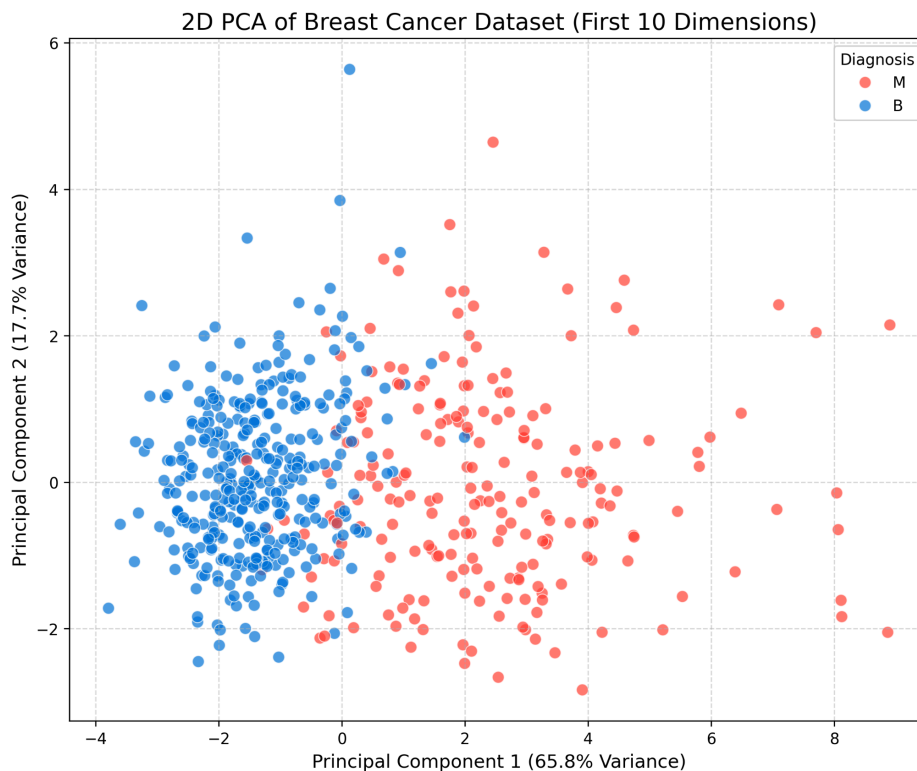


Figure 8: PCA 2D of Breast Cancer Dataset

Analysis of Results

Clear Class Separation

The plot reveals two distinct clusters corresponding to Benign (Blue) and Malignant (Red) tumors. The separation is visually apparent, with minimal overlap between the two classes.

Linear Separability

While there is some overlap in the transition zone, the two classes are largely linearly separable. This indicates that even simple linear classifiers (e.g., Logistic Regression, Linear SVM) could achieve high accuracy on this reduced 2D representation.

Variance Distribution

PC1 (X-axis, 65.8% variance): Captures the primary differentiator between tumor types. This component likely represents a combination of size-related features (radius, area, perimeter) and compactness measures.

PC2 (Y-axis, 17.7% variance): Adds secondary discrimination, possibly capturing texture and shape irregularity.

The majority of class separation occurs along PC1, suggesting that tumor size and compactness are the dominant factors distinguishing malignant from benign cases.